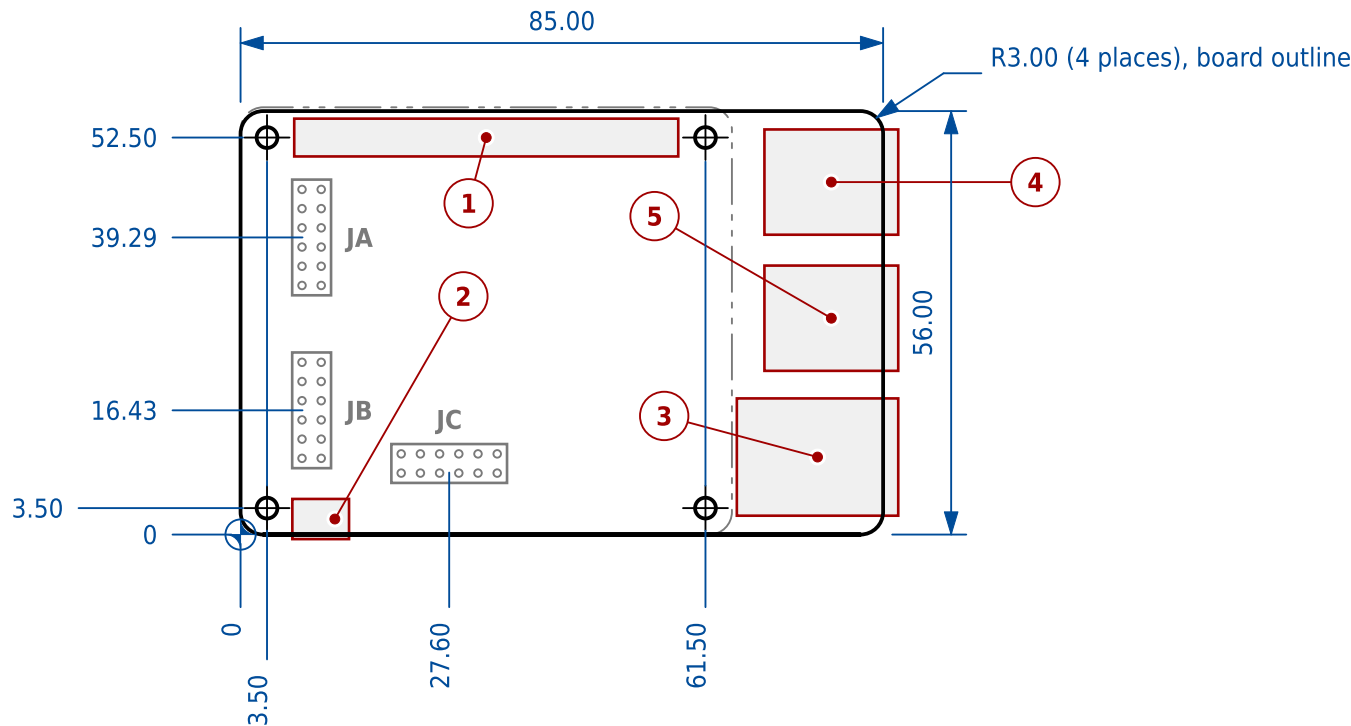




### NOTES

1. All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
2. Phantom outline is the Digilent Pmod HAT Adapter on the 40-pin GPIO header, drawn in this board's frame: its mounting holes coincide with this board's.
3. Connector outlines are the component body as drawn by Raspberry Pi Ltd, overhang past the board edge included.
4. Hole IDs are assigned by this drawing, bottom row first then left to right, and mean the same thing on every Raspberry Pi sheet here.
5. This sheet covers the Raspberry Pi 3 Model B and the Raspberry Pi 3 Model B+. Every connector position was read from both drawings separately and required to match, so the sheet covers both only while they agree.
6. The 40-pin GPIO header is fixed by the Raspberry Pi HAT specification and carries the same position on every Raspberry Pi sheet here. The drawings were read separately, agreed to 0.060 mm, and were snapped to one value.
7. Design mounting hole keep-outs to 6.2 mm diameter on every model: that is what the Raspberry Pi HAT specification (github.com/raspberrypi/hats) requires, and the largest figure the models themselves publish. The KEEPOUT column gives what this model's drawing shows, which is not always anything.
8. The hole diameter tolerance in the schedule, +/-0.05 mm, is quoted from this model's own drawing and governs in place of the hole diameter figure in the general tolerance block.
9. Hole positions are the 3.5 mm inset and 58 x 49 mm rectangle dimensioned on this model's drawing.

### NOTES (continued)

10. Pmod host JC on the Pmod HAT Adapter overhangs the lower board edge, where these boards carry their HDMI and power connectors. Digilent's manual warns to fit the two standoffs opposite the 40-pin connector so JC's pins cannot touch the HDMI shell. The HDMI connectors are not drawn here.
11. Those host positions are DERIVED, not published: Digilent issue no mechanical drawing for the adapter. Good to about +/-0.75 mm. JA and JB face out of the left edge, JC out of the lower edge. The Pmod HAT Adapter sheet has the derivation.
12. Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
13. Assembled envelope, connector overhang included, is 87.00 x 56.60 mm against a 85.00 x 56.00 mm board outline. The FEATURE SCHEDULE gives the extents.
14. GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states. A tolerance quoted in the hole schedule is from the source drawing and governs.
15. Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

### SOURCES

1. Mechanical drawing: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.dxf> - Raspberry Pi Ltd

### HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm       | KEEPOUT mm |
|-----|-------|-------|--------------|------------|
| MT1 | 3.50  | 3.50  | 2.75 +/-0.05 | not given  |
| MT2 | 61.50 | 3.50  | 2.75 +/-0.05 | not given  |
| MT3 | 3.50  | 52.50 | 2.75 +/-0.05 | not given  |
| MT4 | 61.50 | 52.50 | 2.75 +/-0.05 | not given  |

### FEATURE SCHEDULE

| # | FEATURE                     | X EXTENT mm    | Y EXTENT mm    |
|---|-----------------------------|----------------|----------------|
| 1 | 40-pin GPIO header          | 7.10 to 57.90  | 50.00 to 55.00 |
| 2 | micro-USB power input       | 6.85 to 14.35  | -0.60 to 4.71  |
| 3 | Ethernet RJ45               | 65.65 to 87.00 | 2.50 to 18.00  |
| 4 | USB 2.0 type A (upper pair) | 69.30 to 87.00 | 39.65 to 53.57 |
| 5 | USB 2.0 type A (lower pair) | 69.30 to 87.00 | 21.65 to 35.57 |

### PMOD HAT ADAPTER HOSTS

| PORT | EDGE        | CX mm  | CY mm  | PIN1 X mm | PIN1 Y mm |
|------|-------------|--------|--------|-----------|-----------|
| JA   | left edge   | 9.400  | 39.290 | 8.130     | 45.640    |
| JB   | left edge   | 9.400  | 16.430 | 8.130     | 22.780    |
| JC   | bottom edge | 27.600 | 9.400  | 21.250    | 8.130     |

### LEGEND

- Board outline
- Component body, scheduled
- Adjacent part
- Dimension, extension and leader

### SOURCES (continued)

2. Mechanical drawing: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-plus-mechanical-drawing.dxf> - Raspberry Pi Ltd, Raspberry Pi 3 Model B+
3. Mounting holes: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.pdf> - Drawing note on the Model B drawing, quoted: 4x M2.5 MOUNTING HOLES DRILLED TO 2.75 +/- 0.05mm. The B+ drawing gives no hole size, but its geometry is identical.

## RASPBERRY PI 3 MODEL B AND B+

85 x 56 mm, both models

|                           |                  |                                                     |       |
|---------------------------|------------------|-----------------------------------------------------|-------|
| DRAWN                     | VERSION          | UNITS                                               | SCALE |
| generated                 | v0.0-99-g5669bc4 | mm                                                  | 1:1   |
| SIZE                      | SHEET            | DRAWING NO                                          | REV   |
| A3                        | 1 OF 1           | RPI-01                                              | A     |
| MATERIAL                  |                  | GENERAL TOLERANCE                                   |       |
| PCB, thickness not stated |                  | edge +/-0.20 hole pos +/-0.10 hole dia per schedule |       |